INFO33921 Review for the Mid-Term Exam

1. The classics:

a) What is the decryption function for the affine cipher $f(M) = (7M + 10) \bmod 32$?

b) What is the inverse permutation for $\{6,5,1,2,4,3\}$?

c) What is the decryption keyword for the Vigenere cipher with keyword "exam"?

2. Multiplicative inverses; Fermat, Euclid, Euler, and Galois:

a) How many possible private keys, $\{a,b\}$, are there for $f(M) = (aM + b) \bmod 32$?

b) Find a multiplicative inverse for 13 in $\mathbb{Z}_{37}$ using Fermat's Little Theorem.

c) Find a multiplicative inverse for 13 in $\mathbb{Z}_{37}$ using the Extended Euclidean Algorithm.

d) Find a multiplicative inverse for 2 in $\mathbb{Z}_{91}$ using Euler's Theorem.

e) Find a multiplicative inverse of (11) in GF(2^2) with $m(x) = x^2 + x + 1$ using trial and error.

3. RSA cryptosystem; public key {*n*=133, *e*=17}, block size of 3-digits:

a) Encrypt the plaintext "101 006 014".

b) Compute the decryption exponent *d*.

c) Sign the message "109".

4. Diffie-Hellman key exchange protocol, {*p*=23, *g*=5}. A sends B the value 10. B sends A the value 18. What is the shared secret key?

5. Evaluate in GF(2^3) with $m(x) = x^3 + x + 1$ using both <u>polynomial math</u> AND <u>3-bit binary</u>:

(101) + (110)   (101)$\times$(110)

6. Evaluate in GF(2^8) with $m(x) = x^8 + x^4 + x^3 + x + 1$ using <u>8-bit binary</u>:

$\{\text{AF}\} + \{99\}$ $\{9\text{C}\} \times \{02\}$ $\{85\} \times \{03\}$ $\{93\} \times \{42\}$

7. Compute the first ten bits of the binary keystream generated by the 3-bit LFSR with constants (1,0,1) and initial bits (001).

8. Calculate the *initial permutation of S*, and the first three elements in the keystream of a "mini" RC4 with n=4 and key=[1,3,2].

9. Encrypt "*over*" using $\mathbb{Z}_{26}$ encoding and the Hill Cipher with $K = \begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix}$.

10. The current state of an 8-bit Feistel cipher is 11001010. Compute the next state using

$f(R^{i-1}, K^i) = P(R^{i-1} \oplus K^i)$, where $P = \{4,3,2,1\}$, with round key 1001.

Answers:

1.

a) f(C)=23C+26
b) {3,4,6,5,2,1}
c) "wdao"

2.

a) 512
b) 13^{35} mod 37
c) 20
d) 2^{71} mod 91 = 46
e) (10)

3.

a) "054 111 091"
b) 89
c) 81x130x23x109mod133= "072"

4. 13

5. 011, 011

6. {36}, {23}, {94}, {DC}

7. "0011101001…"

8. S=[3,1,0,2]
"00 00 11…"

9. "*jzrv*"

10. 10100000